DocuMind AI - Technical Architecture Report

Page 1: System Overview and Core Design

Project Name: DocuMind AI - RAG-Based Document Assistant.

Author: Software Development Engineering Intern Portfolio.

System Goal: Enable conversational question answering over private PDF documents using Retrieval-Augmented Generation (RAG).

Architecture Overview: The system consists of a React Vite frontend and an Express Node.js backend. When a PDF is uploaded, text is extracted while preserving exact page numbers.

Document Chunking: The extracted text is split into semantic chunks with a configurable chunk size of 600 characters and an overlap of 120 characters to prevent loss of context across boundaries.

Vector Embeddings: Each chunk is converted into high-dimensional vector representations using embedding model APIs such as Google Gemini text-embedding-004 or OpenAI text-embedding-3-small.

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Page 2: Vector Search and Similarity Metric

Vector Storage: Embeddings are indexed in an in-memory vector store equipped with metadata tracking.

Similarity Calculation: The vector store calculates Cosine Similarity between the user question vector and all document chunk vectors.

Cosine Formula: Cosine similarity measures the cosine of the angle between two multi-dimensional vectors, yielding a score between 0.0 and 1.0.

Retrieval Strategy: The top 4 most relevant chunks with highest cosine similarity are retrieved as context for the language model.

Grounding Constraint: The prompt instructs the LLM to strictly base answers on retrieved excerpts and cite specific page numbers.

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Page 3: Evaluation, Guardrails, and Performance

Hallucination Guardrails: If retrieved chunks do not contain enough facts to answer a question, the assistant explicitly states: Insufficient information was found in the uploaded document.

Source Citation: Each response includes clickable citation cards with exact page numbers and similarity confidence scores.

Security Standards: Secret API keys are stored strictly in backend environment variables and never exposed to the frontend browser.

Performance Benchmarks: Average document processing time is under 1.5 seconds for typical 10-page documents.

Internship Conclusion: The project demonstrates full-stack proficiency, prompt engineering, vector search implementation, and clean architectural design.